

Training Report

Comprehensive analysis of model performance and deployment readiness.

Training Analytics Detail

Comprehensive analysis of the training process and model performance

Dataset Details

Information about the data used to train and validate this model.

Item	Value	What it means
Training Epochs	2	The AI studied the data 2 times to learn.
Validation Images	70	Number of images used to test the AI after training.
Total Instances	70	Total number of labeled objects in the validation set.
Image Size	640px	All images were resized to 640×640 pixels.
Classes	2	dog, cat

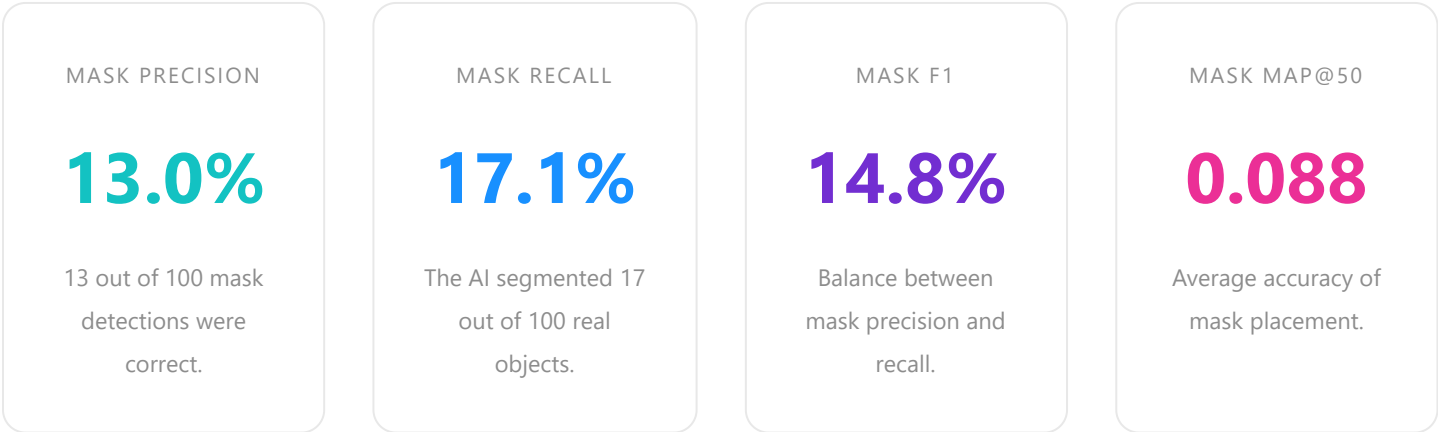
Box Detection Metrics

How well the AI draws boxes around objects. Higher values = better performance.

PRECISION 15.6% 16 out of 100 detections were correct.	RECALL 11.1% The AI found 11 out of 100 real objects.	F1 SCORE 13.0% Balance between precision and recall.	MAP@50 0.102 Average accuracy of box placement.
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🎯 Mask Segmentation Metrics

How well the AI draws pixel-perfect masks around objects.



✅ Class-wise Performance

How well the AI performs for each type of object it was trained to detect.

Class	Box Precision	Box Recall	Box F1	Box mAP@50	Mask Precision	Mask Recall	Mask F1	Mask mAP@50
dog	31.3%	22.1%	25.9%	0.203	26.1%	34.3%	29.6%	0.177
cat	0.0%	0.0%	0.0%	0.000	0.0%	0.0%	0.0%	0.000

🎯 Section 02: Prediction Analytics

TOTAL DETECTIONS

379

Across all images

AVG. CONFIDENCE

58.4%

Mean confidence score

GT COVERAGE

280

Ground truth objects available

PERFORMANCE METRICS (VS GROUND TRUTH)

PRECISION

39.1%

Correctness

RECALL

52.9%

Completeness

F1 SCORE

44.9%

Harmonic Mean

TRUE POSITIVE

148

Correct Detections

FALSE POSITIVES

231

Extra Detections

MISSING (FN)

132

Undetected Objects

Class Distribution

Class Name	Detections	Distribution
defect	351	92.6%
dog	28	7.4%

Confidence Range Breakdown

Confidence Range	Total Detections	Distribution
0.0-0.2	0	0.0%
0.2-0.5	84	22.2%
0.5-0.8	295	77.8%
0.8-1.0	0	0.0%

Section 03: Evidence-Based Diagnostics

CRITICAL ADJUSTMENTS REQUIRED



The model has significant blind spots that make it unsafe for autonomous operation. Retraining with targeted data is mandatory.

Deep-dive analysis of model failures by object size and spatial location.

3.1 Size Performance (Scale Fidelity)

Critical Gaps in Size Detection

OBSERVATIONS:

- The model is struggling to find Tiny objects, achieving only 44.4% reliability.
- Specifically, the class "defect" is getting missed frequently at this size.
- The model is struggling to find Large objects, achieving only 38.3% reliability.
- Specifically, the class "defect" is getting missed frequently at this size.

💡 RECOMMENDATIONS:

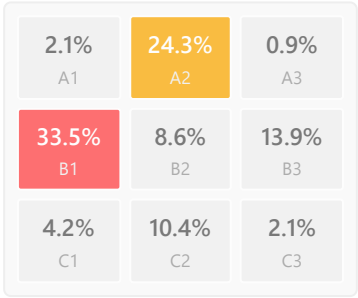
- Pixel Perfection: Tiny items have very few pixels. If your label is even slightly off-center, the model will miss it. Audit these boxes for a tight fit.
- Model Blindness: Items might be too small for the camera's resolution. Add zoomed-in data to show the model the fine details.
- Empty Box Space: Check if your 'Large' boxes have too much background inside. The model learns the wrong shape if the box is loose.
- Data Shortage: You need more pictures where this object is large. The model hasn't seen enough 'Up-close' examples.

Size Group	Reliability AT 46% TARGET	Samples
Tiny	44.4%	63
Small	100%	12
Medium	97.7%	43
— Large	38.3%	162
defect	28.3%	127
dog	74.3%	35

Complex Spatial Blind Spots Identified

OBSERVATIONS:

- Left/Right Imbalance: The left side of the image has 23% more errors than the opposite side. This suggests an environmental factor on that side.
- ⚠ Critical Zone B1 (center-left): 33.5% of all failures — 35 missed objects and 78 false alarms. Most affected class: "defect".
- Watch Zone A2 (top-center): 24.3% error density — 49 missed objects and 33 false alarms. Class: "defect".
- Size Breakdown: Tiny: 96 errors (mixed errors: 35 missed, 61 false alarms) — danger zones: A2 (top-center, 44.8%), B3 (center-right, 39.6%) · Small: 89 errors (mostly false alarms: 0 missed, 89 false alarms) — danger zone: B1 (center-left, 56.2%) · Medium: 28 errors (mostly false alarms: 1 missed, 27 false alarms) — danger zones: A2 (top-center, 39.3%), B1 (center-left, 32.1%) · Large: 124 errors (mostly missed objects: 100 missed, 24 false alarms) — danger zone: B1 (center-left, 41.1%).
- Class Breakdown: "defect": 328 errors (mixed: 127 missed, 201 false alarms) · "dog": 9 errors (mostly missed: 9 missed, 0 false alarms).



ERROR DENSITY HEATMAP

💡 RECOMMENDATIONS:

- Verify Lighting: Check if persistent shadows, glares, or obstructions exist mainly on the left side of the capture area.
- Priority Audit (B1): Manually review "defect" labels in the center-left area of your images. Focus on over-sensitive detections.

Section 04: Deployment Readiness

Automated stability audit to determine if the model is ready for live production use.

RELIABILITY LIMIT

56%

Maximum Confidence Score

OPTIMAL BALANCE POINT

46%

Recommended Confidence Setting

AUTOMATION RATE

51%

% of real objects found by model

DEPLOYMENT STATUS

NOT READY

Not Ready for Production.
Retraining Required.

⚠ WHY NOT READY:

- F1 score (46.1%) is below 60% minimum threshold
- MAXIMUM SAFE CONFIDENCE (56%) is below 60% stability threshold
- False alarms (201) exceed correct detections (144)

🔍 PRODUCTION ASSESSMENT

- [CEILING] MAXIMUM SAFE CONFIDENCE at ~56%: TP collapses beyond this point.
- [OPTIMAL] 46% confidence (Best F1 Score).
- [RESULTS] 144 correct, 201 false alarms, 136 missed (F1=46.1%).
- [WHY NOT 45%] FP rises faster than TP (more false alarms than value).
- [WHY NOT 47%] FN increases and Automation drops (more missed objects).

↑ TRAINING ASSESSMENT

- [CEILING] MAXIMUM SAFE CONFIDENCE at ~56% (structural drop zone).
- [TARGET] Pre-wall confidence: 45% (best usable region before collapse).
- [AT 45%] TP=148, FP=231, FN=132 → Automation=53%.
- [IoU] Target IoU: 0.80 (box quality goal for next training).
- [DIAGNOSIS] Improve confidence strength to push ceiling right. Model needs stronger feature extraction.